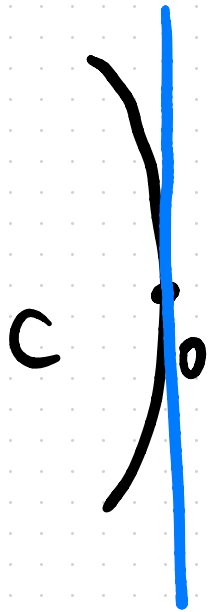


Lemma: Let C be a convex cone which is not all of V . Then there is a hyperplane supporting C at 0 .

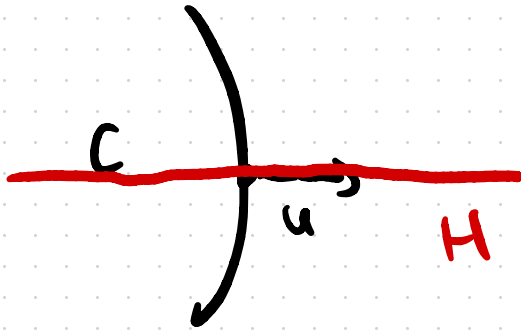


Proof: Induction on $\dim V$.

If $\dim V \leq 2$, exercise.

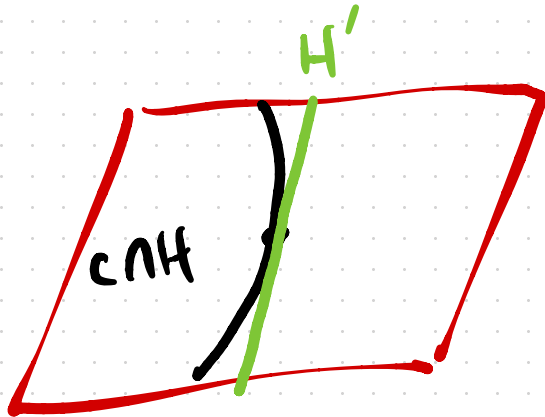
Assume $\dim V \geq 3$. Let $u \in V \setminus C$.

Let H be a hyperplane through u .



$C \cap H$ is a cone in H which is not all of H .

By the inductive hypothesis, there is a hyperplane H' of H which supports $C \cap H$ at 0 .



Consider the quotient space V/H' .

We have $\dim V/H' = 2$

$q: V \rightarrow V/H'$ is the quotient map

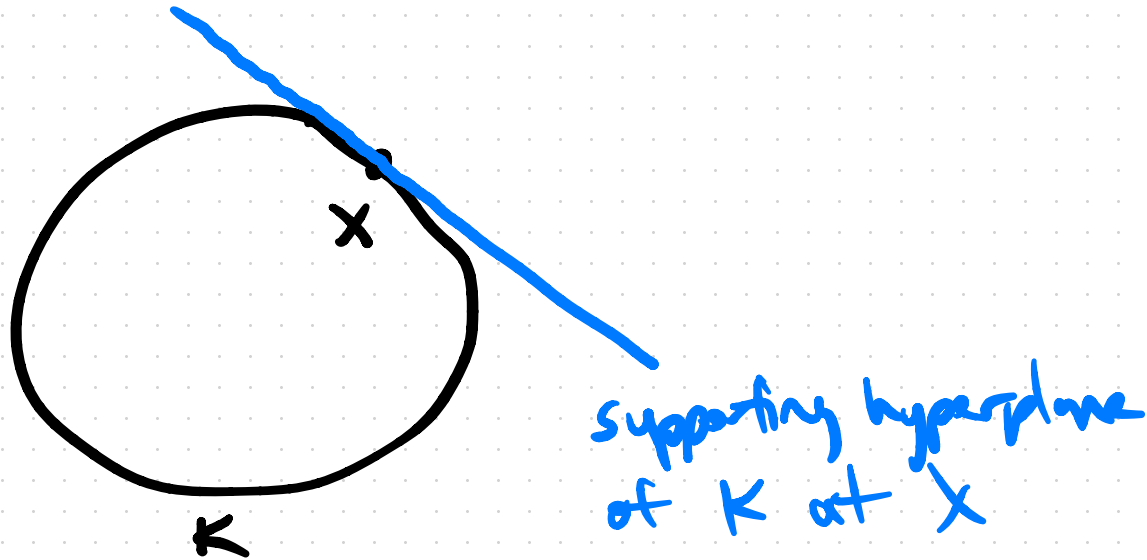
V/H'
 $q(C)$ }
 H''

$q(C)$ is a cone in V/H' , and is not the entire space V/H' .

Therefore, by the inductive hypothesis, there is some line H'' supporting $q(C)$ at 0 .

Then $q^{-1}(H'')$ is a hyperplane supporting C at 0 .

Supporting hyperplane theorem: Let K be a closed convex set, and let x be on the boundary of K . Then there is a hyperplane passing through x so that K is weakly on one side of the hyperplane.



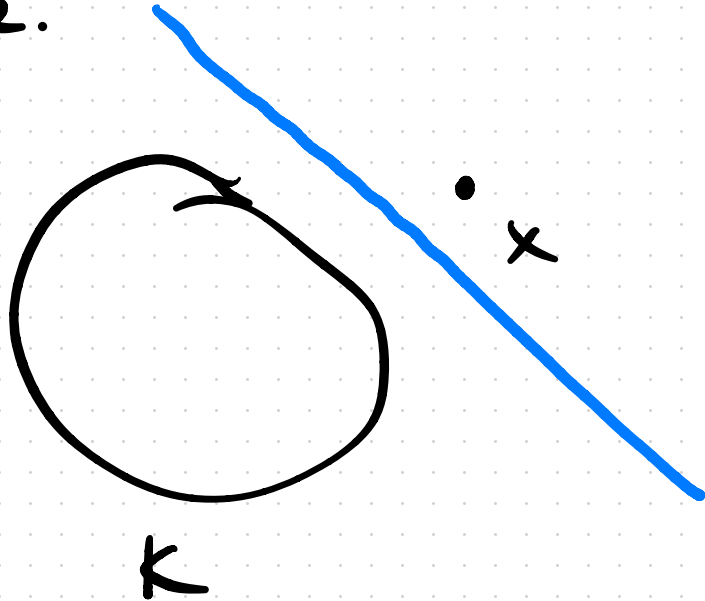
Let the **direction cone** of K at x be

$$D(K, x) = \text{cone}(K - x)$$

$$= \text{cone} \{ u - x : u \in K \}$$

Because x is on the boundary of K , $D(K, x)$ is not the entire space. Therefore, $D(K, x)$ has a supporting hyperplane at 0 . Shifting this by x gives a supporting hyperplane of K at x .

Separating hyperplane theorem: Let K be a closed convex set, and let $x \notin K$. Then there is a hyperplane such that x and K are strictly on opposite sides of the hyperplane.

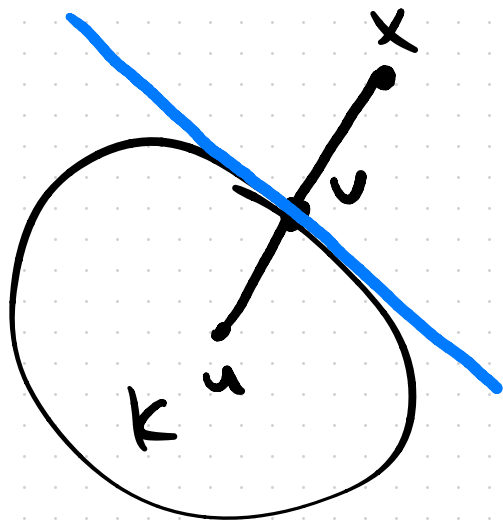


Proof: First assume $x \notin \text{aff } K$. Then we can find
a hyperplane containing K and not x . Shifting this
hyperplane gives a separating hyperplane.

$x \cdot$



Now assume $x \notin \text{aff } K$. By restricting to $\text{aff } K$, we can assume $V = \text{aff } K$. Let u be a point in the interior of K . Then



the line segment $[u, x]$ contains a boundary point v of K .

$v \neq u, x$, since K is closed.

We apply the supporting hyperplane theorem at v . Shifting this hyperplane gives a separating hyperplane.

Corollary: Let K be a closed convex set. Then

$$K = \{x \in V : y(x) \leq \sup_K y \text{ for all } y \in V^*\}$$

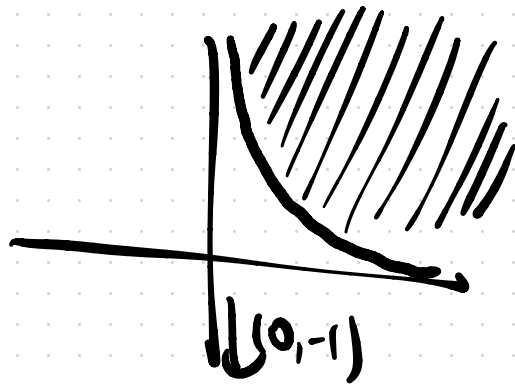
Proof: Clearly $K \subseteq \dots$

If $x_0 \notin K$, then there is a

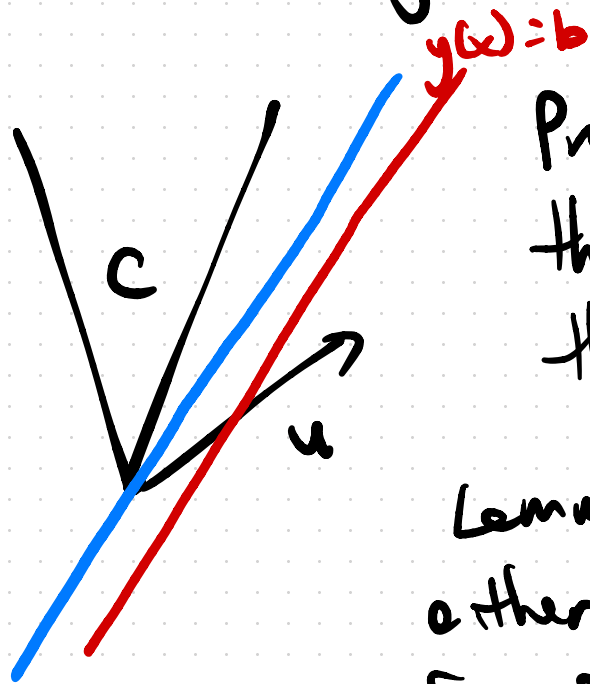
hyperplane $\{y(x) = b\}$ separating

x_0 and K . Assume $y(x) \leq b$ for all $x \in K$.

Therefore $y(x_0) > b \geq \sup_K y$. Thus $x_0 \notin \dots$



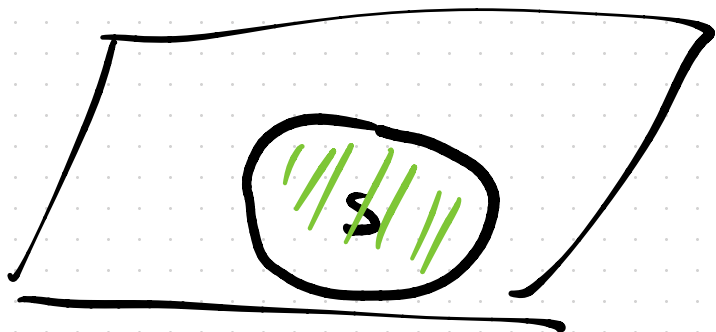
Corollary (Farkas' Lemma): Let C be a closed convex cone and $u \notin C$. Then there exists $y \in V^*$ such that $y(u) > 0$ and $y(x) \leq 0$ for all $x \in C$.



Proof: Apply hyperplane separation theorem to C and u . Shift the resulting hyperplane to 0.

Lemma: For any $y \in V^*$ and cone C , either y achieves a maximum of 0 on C , or it is unbounded from above on C .

Given any set $S \subset V$, the **relative interior** of S , **$\text{rint } S$** , is the interior of S as a subset of $\text{aff } S$.



$$\dim \text{aff } S = 2$$

$$\begin{aligned} \text{int } S &= \emptyset \\ \text{rint } S &= \text{green lines} \end{aligned}$$

$$V = \mathbb{R}^3$$

Given a set $S \subset V$, the **lineality space** of S is

$$\text{lin } S = \begin{cases} \{v \in V : x + \lambda v \in S, x \in S, \lambda \in \mathbb{R}\} & \text{if } S \neq \emptyset \\ \{0\} & \text{if } S = \emptyset \end{cases}$$

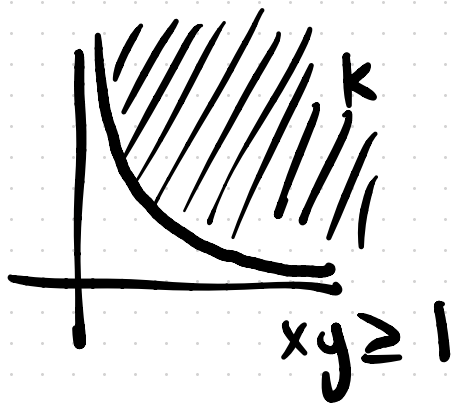
The **recession cone** of S is

$$\text{rec } S = \begin{cases} \{v \in V : x + \lambda v \in S, x \in S, \lambda \geq 0\} & \text{if } S \neq \emptyset \\ \{0\} & \text{if } S = \emptyset \end{cases}$$

If K is convex and nonempty

$$\text{lin } K = \{v \in V : x + \lambda v \in K \text{ for some } x \in K \text{ and all } \lambda \in \mathbb{R}\}$$

$$\text{rec } K = \{v \in V : x + \lambda v \in K \text{ for some } x \in K \text{ and all } \lambda \geq 0\}$$



$$\text{lin } K = \{0\}$$

$$\text{rec } K = \{x \geq 0, y \geq 0\}$$

If K is bounded, then $\text{link} K = \text{rec} K = \{0\}$.

Let $K \subset V$ be a convex set. A **face** of K is a nonempty convex subset $F \subset K$ such that if $x \in F$ and x can be written as a positive convex combination of $x_1, \dots, x_m \in K$, then $x_1, \dots, x_m \in F$.

For cones, "convex combination" can be replaced by "conical combination".

A **support set** of a convex set K is a nonempty set of the form $\operatorname{argmax}_K y$, where $y \in V^*$.

An **extreme ray** of a convex cone is a face of the cone which is a ray.

Thm: Every closed convex cone with lineality space $\{0\}$ is the conical hull of its extreme rays.

